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MILITARY SPECIFICATION

COUPLERS, PASSIVE, FIBER OPTIC, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24621. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic couplers for use in Navy shipboard systems. The couplers (see 6.6.2) are used for coupling and splitting optical signal power, including optical power in wavelength-division multiplexed (WDM) (see 6.6.13) signals.

1.2 Classification. Couplers are classified as specified in 1.2.1 through 1.2.3 (see 6.2).

1.2.1 Type. The type designation defines the wavelength of operation of the coupler.

Type 1 - Nominal center of operation at 1.31 micrometers (μm), with a range of 1.26 to 1.36 μm.
Type 2 - Nominal center of operation at 1.55 μm, with a range of 1.50 to 1.60 μm.
Type 3 - Wavelength dependent (WDM devices).
Type 4 - Wavelength independent (range of operation that includes part or all of the wavelength ranges of types 1 and 2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Class. The class designation defines the optical power distribution made by the coupler. The tolerances of the elements of the transmittance matrix (see 6.6.3) that allow the coupler to be classified as uniformly distributive or nonuniformly distributive (see 6.6.6 and 6.6.12) will be indicated in the specification sheet (see 3.1).

Class A - Uniformly distributive.
Class B - Nonuniformly distributive.

1.2.3 Style. The style designation defines the style of packaging.

Style 1 - Printed circuit board (PCB) mounted coupler with multimode pigtails.
Style 2 - PCB-mounted coupler with single-mode pigtails.
Style 3 - Stand-alone coupler with multimode pigtails.
Style 4 - Stand-alone coupler with single-mode pigtails.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-I-735	- Isopropyl Alcohol.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-55330	- Connectors, Electrical and Fiber Optic, Packaging of.
MIL-C-85045	- Cables, Fiber Optics, (Metric) General Specification for.

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STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1344 - Test Methods for Electrical Connectors.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-2 - Impact Test Measurements for Fiber Optic Devices. (DoD adopted)
455-4 - Fiber Optic Connector/Component Temperature Life. (DoD adopted)
455-5 - Humidity Test Procedure for Fiber Optic Connecting Devices. (DoD adopted)
455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DoD adopted)
455-12 - Fluid Immersion Test Procedure for Fiber Optic Connecting Devices.
455-16 - Salt Spray (Corrosion) Test for Fiber Optic Components.
455-20 - Measurement of Change in Optical Transmittance. (DoD adopted).
455-22 - Ambient Light Susceptibility of Fiber Optic Components. (DoD adopted)
455-34 - Interconnection Device Insertion Loss Test. (DoD adopted)
455-36 - Twist Test for Fiber Optic Connecting Devices. (DoD adopted)

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Materials. Couplers shall be constructed of material as specified (see 3.1 and herein). Materials selected for coupler usage shall be of a type and quality to ensure compliance with the requirements of this specification, and shall be physically and chemically compatible for their intended lifetime. Unless otherwise specified (see 3.1), combinations of materials used shall be nonmagnetic and non-nutrient to fungus. Materials may be dielectric or electrically conductive as applicable. Where new or questionable material is being considered for use, the manufacturer shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use. Materials used in the coupler construction shall not emit toxic or explosive fumes when exposed to high temperature or flame. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.3.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.3.2 Metals. Unless otherwise specified (see 3.1), all metals shall be nonmagnetic and corrosion-resistant. Dissimilar metals shall not be used in intimate contact with each other unless finished to prevent electrolytic corrosion. The criteria for the selection and protection of dissimilar metal combinations shall be in accordance with the requirements of MIL-STD-889.

3.3.3 Nonmetallic materials. Nonmetallic materials used in the construction of couplers shall not be degraded when exposed to the specified environmental conditions.

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3.3.4 Sealing compounds. Sealing compounds shall not flow at the maximum specified storage temperature nor crack at the minimum specified storage temperature.

3.3.5 Solvents, adhesives, and cleaning agents. If any chemical or cements are used in bonding of internal coupler parts or in the fiber/contact securing-polishing processes, incompatibility shall not exist between the materials employed such that degradation of these materials results from in-service use. Any epoxy and adhesives used in coupler construction shall conform to MIL-STD-454, requirement 23.

3.3.6 Material Safety Data Sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.2).

3.4 Design and construction. The coupler shall be of the design and construction as specified (see 3.1). The coupler shall include optical pigtails for interconnection within an optical link as input and output ports, an optical coupling junction, and a protective housing. The port configuration of the input and output ports shall be as specified (see 3.1). The coupler shall provide reliable operation under the environmental and mechanical requirements specified herein for the intended lifetime of the coupler which shall be not less than 20 years.

3.4.1 Fiber optic pigtail. The coupler shall be supplied with an optical fiber or optical cable pigtail (see 6.6.7, 6.6.8) as specified (see 3.1). One end of the pigtail shall be permanently mounted in the coupler housing. The mounting construction shall provide strain relief for the pigtail. The pigtail length shall be not less than 1 meter. Metal components shall not be used within the pigtail, except for mounting to the coupler housing or termination connector.

3.4.2 Housing. The coupling junction shall be enclosed in a sealed watertight housing as specified herein. The housing for couplers shall include hardware for mounting.

3.4.3 Dimensions. The coupler shall be enclosed in a physical envelope with dimensions and tolerances as specified (see 3.1). The physical envelope shall include the coupler housing and fiber optic pigtails (see 4.6.1.1).

3.4.4 Weight. The coupler weight limitations (see 4.6.2) shall be as specified (see 3.1).

3.4.5 Color and finish. The color and finish of the coupler housing shall be as specified (see 3.1). Unless otherwise specified (see 3.1), the optical pigtail color shall be natural (see 4.6.1.1).

3.4.6 Maintainability. The coupler shall require no preventive maintenance.

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3.4.7 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.5 Performance. The performance requirements shall be defined in terms of optical, mechanical, environmental, and chemical properties.

3.5.1 Optical properties.

3.5.1.1 Optical transmittance. Each P_{ij} of the coupler transmittance matrix (see 6.6.3) associated with each specified wavelength range shall be as specified (see 3.1 and 4.6.3.1). The values, P_{ij}, of the transmittance matrix define all of the port-to-port characteristics for the coupler.

3.5.1.1.1 Isolation. Unless otherwise specified (see 3.1), the optical power coupled to any isolated port (see 6.6.5) shall be not greater than minus 35 decibels (dB) (see 4.6.3.1).

3.5.1.1.2 Backscatter. The backscatter (see 6.6.1) of any port shall be not greater than minus 30 dB (see 4.6.3.1).

3.5.1.2 Change in optical transmittance. Each P_{ij} of the transmittance matrix associated with one wavelength of operation of the coupler shall not change more than 0.3 dB when subjected to mechanical and environmental stress (see 4.6.3.2). For couplers with more than eight ports total, a minimum of seven ports shall be monitored. Nonadjacent ports shall be used as specified (see 3.1).

3.5.1.3 Wavelength sensitivity. The change in transmittance shall be less than 0.2 percent per nanometer at the wavelength spectrum specified (see 3.1 and 4.6.3.3).

3.5.1.4 Ambient light susceptibility. The optical power from any one port or sum of ports shall be not greater than minus 50 dBm (see 4.6.3.4).

3.5.1.5 Polarization sensitivity (see 6.6.9). The normalized optical transmittance shall not deviate more than 2 percent over 180 degrees of rotation for single-mode couplers (see 4.6.3.5).

3.5.2 Mechanical properties.

3.5.2.1 Impact. The coupler shall not exhibit any evidence of physical damage (see 4.6.4.1).

3.5.2.2 Cable seal flexing. Each pigtail strain relief shall prevent loss of environmental sealing or other damage that may impair the coupler operation (see 4.6.4.2).

3.5.2.3 Crush (stand-alone couplers only). The coupler shall show no evidence of broken parts or accessories, loss of optical continuity, or damage to seals (see 4.6.4.3).

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3.5.2.4 Cable pull-out force. The cable-to-coupler pull-out strength shall be a minimum of 50 percent of the specified tensile strength of the attached cable or 100 newtons, whichever is less. There shall be no cable jacket damage, cable clamp failure, cable-to-coupler seal damage, distortion or bending of coupler parts, or cable disengagement from the coupler (see 4.6.4.4).

3.5.2.5 Housing bending moment. The coupler shall sustain no permanent deformation (see 4.6.4.5).

3.5.2.6 Twist. The coupler and cable assembly shall reveal no loss of environmental sealing, splitting, or other impairment due to the torsional bending exposure (see 4.6.4.6).

3.5.2.7 Fiber dynamic strength. The minimum fiber dynamic strength shall be 50 percent of the specified proof strength of the attached fiber. There shall be no fiber-to-coupler seal damage or distortion or bending of coupler parts (see 4.6.4.7).

3.5.3 Environmental properties. The coupler shall meet all requirements specified (see 3.1) during the specified operating environments and after the specified storage environment. The operating temperature range and storage temperature range shall be as shown in table I, as specified (see 3.1).

TABLE I. Temperature ranges.

Range	Package	Operating (°C)	Storage (°C)
1	Stand-alone	-54 to +65	-62 to +71
2	Stand-alone	-28 to +65	-62 to +71
3	PCB-mounted	-54 to +85	-62 to +85
4	PCB-mounted	-28 to +85	-62 to +85

3.5.3.1 Thermal shock. The coupler shall show no evidence of part dimensional change, softening, surface damage, identification or marking impairment, leakage of waterproofing compounds, loss of sealing capability, or other damage detrimental to the operation of the coupler (see 4.6.5.1).

3.5.3.2 Fluid immersion. The coupler shall reveal no cracks, splits, gumminess, voids, swelling or softening of material, loss of identification marking, discoloration, or other effects detrimental to the operation of the coupler (see 4.6.5.2).

3.5.3.3 Temperature life. The coupler shall not exhibit any visual evidence of dimensional change, opening of seals, cracking or crazing of components, or any form of physical damage (see 4.6.5.3).

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3.5.3.4 Temperature-humidity cycling. The coupler shall reveal no component softening, surface damage, or identification marking impairment (see 4.6.5.4).

3.5.3.5 Vibration. The coupler shall exhibit no evidence of physical damage (see 4.6.5.5).

3.5.3.6 Shock. The coupler shall exhibit no evidence of physical damage (see 4.6.5.6).

3.5.3.7 Salt spray. No visible evidence of corrosion shall be seen on the external or associated coupler parts (see 4.6.5.7).

3.5.3.8 Water pressure (watertight couplers only). The coupler shall reveal no evidence of penetration of indicator dye into the sealed region of the coupler (see 4.6.5.8).

3.5.3.9 Nuclear radiation resistance. Unless otherwise specified (see 3.1), the coupler shall meet the requirements for gamma radiation as specified (see 3.1 and 4.6.5.9).

3.5.4 Chemical properties.

3.5.4.1 Fungus resistance. Materials used in the construction of the coupler shall meet the requirements of MIL-STD-454, requirement 4 for fungus-inert materials. Materials not identified in MIL-STD-454, requirement 4 as fungus inert shall meet the grade I classification of MIL-STD-810, method 508 (see 4.6.6.1).

3.5.4.2 Flammability. Flames shall extinguish within the time specified in MIL-STD-1344, condition B of method 1012. Any observed afterglow shall cease within the time specified in MIL-STD-1344, condition B of method 1012. There shall be no dripping that will cause ignition of flammable material placed below the flame's contact point with the coupler. There shall be no violent burning or explosive fire (see 4.6.6.2).

3.6 Transportability. The coupler operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.7 Identification marking. The coupler shall be marked with the assigned part number using the numbering scheme specified (see 6.8). The coupler shall be marked with the manufacturer's name or code, the date of manufacture (year and month), and optical pigtail numbers. The coupler shall be marked to identify the input and output port configuration using the numerical value for input ports followed by a separator and a numerical value for the output ports. An example of proper port identification is 2 x 4 where "2" is the number of input ports and "4" is the number of output ports. Coupler port numbers shall be placed on the coupler housing, beginning with the input ports, a "1" being the first port for stand-alone couplers. On couplers with cable pigtails, the port numbers shall be placed on the pigtail cable.

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3.8 Workmanship. Couplers shall meet all physical and interconnection requirements of this specification. Couplers having damaged parts, loose fibers, cracked or abused surfaces, illegible markings, or inconsistencies in finish or appearance shall be deemed as inferior quality and will be considered adequate basis for rejection during inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.2.1 Toxicological data and formulations. The contractor shall have toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.3 Inspection conditions. Inspection conditions shall be as specified in the applicable test method for each inspection parameter, and as specified (see 3.1).

4.4 First article inspection. First article inspection shall be in accordance with table II and shall be performed on sample units produced with equipment and procedures used in normal production.

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TABLE II. First article inspection.

Inspection	Requirement	Test	Sample size
Group I			
Visual and mechanical	3.4, 3.8	4.6.1.1	8
Dimensions	3.4.3	4.6.1.1	1/
Weight	3.4.4	4.6.2	1/
Color and finish	3.4.5	4.6.1.1	1/
Identification marking	3.7	4.6.1.1	1/
Optical transmittance	3.5.1.1	4.6.3.1	1/
Wavelength sensitivity	3.5.1.3	4.6.3.3	1/
Ambient light susceptibility	3.5.1.4	4.6.3.4	1/
Polarization sensitivity	3.5.1.5	4.6.3.5	1/
Group II			
Thermal shock	3.5.3.1	4.6.5.1	1
Fluid immersion	3.5.3.2	4.6.5.2	2/
Water pressure	3.5.3.8	4.6.5.8	2/
Temperature life	3.5.3.3	4.6.5.3	1
Temperature-humidity cycling	3.5.3.4	4.6.5.4	3/
Vibration	3.5.3.5	4.6.5.5	3/
Shock	3.5.3.6	4.6.5.6	3/
Salt spray	3.5.3.7	4.6.5.7	3/
Group III			
Impact	3.5.2.1	4.6.4.1	1
Cable seal flexing	3.5.2.2	4.6.4.2	4/
Crush	3.5.2.3	4.6.4.3	4/

See footnotes at end of table.

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TABLE II. First article inspection - Continued.

Inspection	Requirement	Test	Sample size
Cable pull-out force	3.5.2.4	4.6.4.4	4/
Housing bend moment	3.5.2.5	4.6.4.5	4/
Twist	3.5.2.6	4.6.4.6	4/
Fiber dynamic strength	3.5.2.7	4.6.4.7	4/
Group IV			
Fungus resistance	3.5.4.1	4.6.6.1	1
Flammability	3.5.4.2	4.6.6.2	1
Nuclear radiation resistance	3.5.3.9	4.6.5.9	3

1/ The same samples shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the thermal shock inspection.
3/ The same sample shall be used as in the temperature life inspection.
4/ The same sample shall be used as in the impact inspection.

4.4.1 Samples. Unless otherwise specified, the number of sample units to be subjected to first article inspection shall be eight couplers of the same part number. The samples shall be taken at random from a production run and shall be produced with equipment and procedures intended for production of final units by the manufacturer.

4.4.2 Inspection routine. The sample shall be subjected to the first article inspection using the criteria specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided into three groups. The sample units shall then be subjected to the inspection for their particular group (group II, III, or IV), and tested in accordance with table II in the order shown. After completion of sample testing, all sample units shall be resubjected to group I inspection. Any coupler failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant first article acceptance.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the groups A, B, and C inspections (tables III, IV, and V) (see 6.5).

4.5.1 Group A inspection. Group A inspection shall consist of the inspections specified in table III. All applicable inspections shall be performed on each sample unit in the order shown.

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TABLE III. Group A inspection.

Inspection	Requirement	Test
Visual and mechanical	3.4, 3.8	4.6.1.1
Dimensions	3.4.3	4.6.1.1
Weight	3.4.4	4.6.2
Color and finish	3.4.5	4.6.1.1
Identification marking	3.7	4.6.1.1
Optical transmittance	3.5.1.1	4.6.3.1
Wavelength sensitivity	3.5.1.3	4.6.3.3
Ambient light susceptibility	3.5.1.4	4.6.3.4
Polarization sensitivity	3.5.1.5	4.6.3.5

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV, in the order shown. Group B inspection shall be performed on sample units that have passed group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test
Shock	3.5.3.6	4.6.5.6
Thermal shock	3.5.3.1	4.6.5.1
Cable seal flexing	3.5.2.2	4.6.4.2
Cable pull-out force	3.5.2.4	4.6.4.4
Twist	3.5.2.6	4.6.4.6
Fiber dynamic strength	3.5.2.7	4.6.4.7

4.5.3 Group C inspection. Group C inspection shall consist of the inspections specified in table V, in the order shown. Group C inspection shall be performed on sample units that have passed groups A and B inspections.

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TABLE V. Group C inspection.

Inspection	Requirement	Test
Temperature life	3.5.3.3	4.6.5.3
Temperature-humidity cycling	3.5.3.4	4.6.5.4
Water pressure	3.5.3.8	4.6.5.8
Fluid immersion	3.5.3.2	4.6.5.2
Impact	3.5.2.1	4.6.4.1
Vibration	3.5.3.5	4.6.5.5
Salt spray (corrosion)	3.5.3.7	4.6.5.7
Crush	3.5.2.3	4.6.4.3
Housing bending moment	3.5.2.5	4.6.4.5
Fungus resistance	3.5.4.1	4.6.6.1
Flammability	3.5.4.2	4.6.6.2
Nuclear radiation resistance	3.5.3.9	4.6.5.9

4.5.3.1 Disposition of sample units. Sample units that have been subjected to group C inspection shall not be delivered.

4.6 Methods of inspection.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical inspections. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed using 5X magnification to verify that the design, construction, physical characteristics, dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Optical test methods. The use of optical test methods to assess the influence of test exposures on performance of couplers requires that they be connected to lengths of fiber. Unless otherwise specified (see 3.1), all optical measurements shall be performed at 1.310 ± 0.020 μm for type 1 couplers and at 1.550 ± 0.025 μm for type 2 couplers. For coupler types 3 and 4 the measurements shall be performed at the wavelengths as specified (see 3.1). The fiber shall be cabled, that is, the couplers shall be attached to lengths of cable of the type

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specified in MIL-C-85045. The cable lengths shall be not greater than 30 meters. The optical fiber components exposed at the test set (source, detector, or coupler) may be connectorized for access to the test set. For multimode fiber, the source used shall be noncoherent. Light launch conditions shall be in accordance with table VI.

TABLE VI. Light launch conditions.

Fiber type	Launch condition
Single-mode	30 mm diameter mandrel
Multimode	70/100 restricted, or 70/70 restricted (see 6.4.10 and 6.4.11)

Any optical power detection method may be used if the method is sufficiently sensitive to measure the differential optical power levels as specified (see 3.6.1.1), and if the method provides repeatable readings (less than 3 percent variation).

4.6.1.3 Equivalent test methods. The use of equivalent test methods is allowed (see 3.1 and 6.2).

4.6.2 Weight. The coupler weight, including pigtails, shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance. The coupler shall meet the requirements specified in 3.4.4.

4.6.3 Performance.

4.6.3.1 Optical transmittance. The transmittance, P_{ij}, shall be determined by measuring the transmittance between input port i and output port j using method C of EIA 455-34 (see 6.3 and appendix D). The backscatter elements of the transmission matrix shall be measured in accordance with appendix A. The P_{ij} shall be measured by launching a known power level into port i for all i ports of the coupler and measuring the output power of all the j ports. This shall be continued until all i ports of the coupler have been individually connected as input ports to the coupler, unless otherwise specified (see 3.1). Cladding mode stripping shall be used on the input pigtail. Tolerances may be associated with each of the specified P_{ij}. The P_{ij} shall be as specified in 3.5.1.1.

4.6.3.2 Change in optical transmittance. The change in optical transmittance shall be tested in accordance with EIA 455-20 or equivalent (see 4.6.1.3), using a monitor fiber. The change in optical transmittance shall be monitored for all of the output ports of a randomly selected input port. Change in optical transmittance shall be as specified in 3.5.1.2.

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4.6.3.3 Wavelength sensitivity. The operational optical wavelength range shall be tested in conjunction with the testing of the transmittance matrix. For each specified wavelength range, the transmittance matrix shall be tested using a source that spans the wavelength range. The source used shall be randomly polarized. Laser sources shall not be used. The wavelength range shall be scanned in 10 nanometer steps. Change in wavelength shall be as specified in 3.5.1.3.

4.6.3.4 Ambient light susceptibility. The ambient light susceptibility shall be measured in accordance with EIA 455-22, with the exception that the output power in the "on" state shall be referenced to 1 milliwatt. The test temperature shall be 25 degrees Celsius (°C). The optical power shall be as specified in 3.5.1.4.

4.6.3.5 Polarization sensitivity. Polarization sensitivity shall be tested as follows: The optical input and output ports of the coupler under test (see figure 1) shall be connected to short optical cables in which the cladding mode stripping occurs before the input to the coupler and between optical cables in which the cladding mode stripping occurs before the input to the coupler and between the relevant output ports and the detectors. Unused coupler ports are appropriately capped to restrict ambient light acceptance. The nominal wavelength of the source shall be 1.310 μm for type 1 couplers, 1.550 μm for type 2 couplers, and as specified for types 3 and 4 couplers. The spectral width of the source shall be at least 0.080 μm. The $\lambda 2$ retardation plate shall be aligned just prior to the polarization sensitivity test such that an extinction ratio of at least 30 dB is measured. The polarizer shall have an extinction ratio greater than or equal to the extinction ratio of the $\lambda 2$ retardation plate. The polarization sensitivity test shall be performed by rotating the $\lambda 2$ retardation plate through 180 degrees. Data shall be obtained for the change in normalized optical transmittance versus polarization. The normalized optical transmittance shall be calculated as follows:

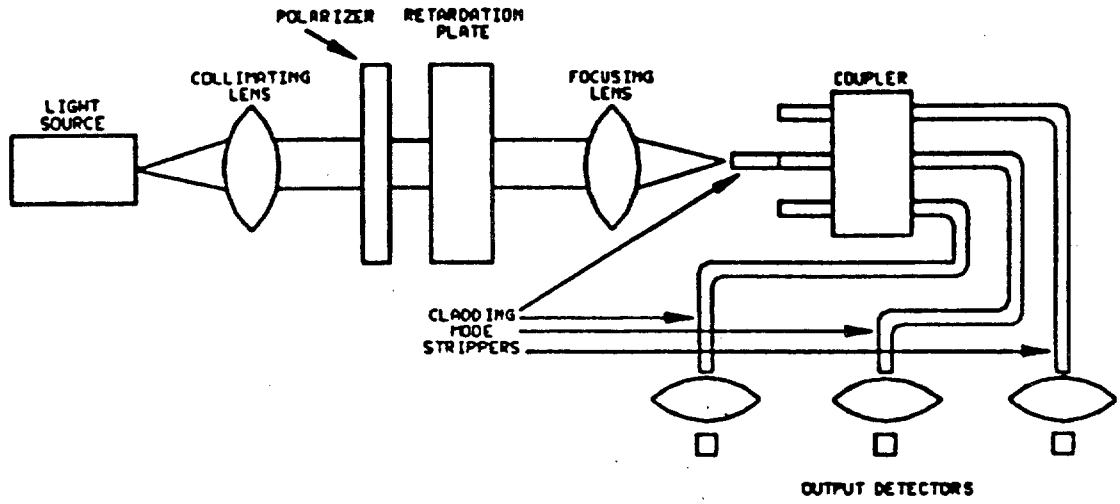
$$P_{norm} = P_{test}/\Sigma P_{output}$$

where P_{test} shall be the output power of the coupler port with the maximum deviation during the test and ΣP_{output} shall be the sum of the powers of the output ports of the coupler. The polarization sensitivity shall be calculated as follows:

$$(P_{norm})_{max} \quad X \quad 100 \text{ percent}$$

where P_{norm} shall be the deviation from the average for the coupler port with the maximum deviation. The requirements of 3.5.1.2 and 3.5.1.5 shall be verified.

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FIGURE 1. Polarization sensitivity test.

4.6.4 Mechanical.

4.6.4.1 Impact. The coupler shall be tested for impact in accordance with EIA 455-2. Stand-alone couplers shall be classified as moderate service class. PCB-mounted couplers shall be classified as light service class. The change in optical transmittance shall be monitored after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1. The coupler shall meet the requirements specified in 3.5.2.1.

4.6.4.2 Cable seal flexing (jacketed fiber pigtail). Couplers with cable pigtails shall be tested for seal flexing in accordance with MIL-STD-1344, method 2017, test procedure I, except that the test sample shall be a coupler. At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1. The coupler shall meet the requirements specified in 3.5.2.2.

4.6.4.3 Crush. Stand-alone couplers shall be tested in accordance with MIL-STD-1344, method 2008, except that the test sample shall be a coupler. The test load shall be 1250 newtons. The change in optical transmittance shall be monitored during and after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.2.3.

4.6.4.4 Cable pull-out force (jacketed fiber pigtail). Couplers with cable pigtails shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied up to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be monitored during and after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.2.4.

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4.6.4.5 Housing bending moment. Couplers shall be tested in accordance with the following procedure that checks proper operation of a coupler under deformation due to applied torque, a condition that can occur as a result of uneven stresses from mounting or installation. This test considers as a reference the practical case of a coupler body 50 millimeters (mm) in length, mounted on a flat surface made uneven by a 0.5 mm shim. This small shim, which is used to model a typical convex surface of attachment in shipyard construction, shall be situated midway underneath the longest extent of the coupler body. The coupler shall be balanced in a stable position atop a small metal block (or shim). The width of the fulcrum (the bar of the shim), upon which the coupler rests, shall be less than or equal to 0.1 mm of the maximum dimension across the top face of the coupler (but not with the contact span less than 2 mm). The fulcrum shall be secured against a smooth, level foundation offering stiff support. Install the coupler according to the manufacturer's recommended mounting instructions, but with the shim centered underneath the coupler. In the event that such instructions are not available from the manufacturer, the sample description below pertains to a coupler bearing four screw holes for attachment to a flat base, as shown on figure 2. The four screws shall be tightened gradually in quarter-turn clockwise increments once the coupler body becomes loosely clamped against the fulcrum during this fastening process. The four screws shall be attended by small quarter-turn twists, one after another, in cyclic permutation such that all four bolts become progressively tighter. The force applied by the mounting screws of the coupler shall be increased until the loading effect deflects the far edges downward to the level of the foundation (but only to the degree that the tips of the coupler body just barely touch the foundation below). This can be determined electronically by sensing the moment that an electrical conduction path is established while the coupler gradually becomes more tightly fastened. The bent coupler is measured at this time and compared with the maximum allowed value, as specified (see 3.1). Each of the four screws (fasteners) shall be tightened. The test configuration and definition of bending moment angle, θ_{BM}, in terms of the deformed coupler body are shown on figure 3. At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1. The coupler shall meet the requirements of 3.5.1.2 and 3.5.2.5.

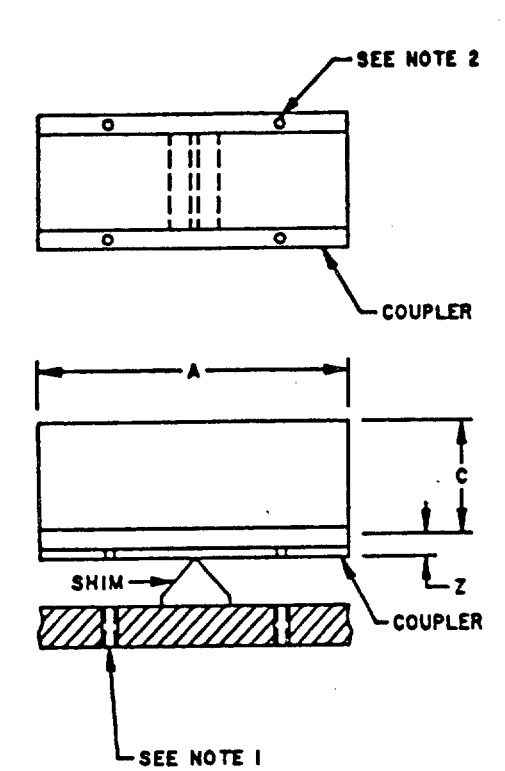
Note: The combination of the shim and coupler, as shown, defines a reference angle (of deformation) of 1.15 degrees. The change in optical transmittance of the deformed coupler under stress/strain is then measured to verify that the change in optical transmittance (including any induced attenuation caused by this test) still conforms to values less than the maximum allowed change in optical transmittance as specified (see 3.1).

Note: The fulcrum bar (or shim) shall extend across the full width of the coupler body. Both the pivot piece of shim (fulcrum bar) and floor support beneath the fulcrum shall be of rigid, inelastic, and smoothly finished material.

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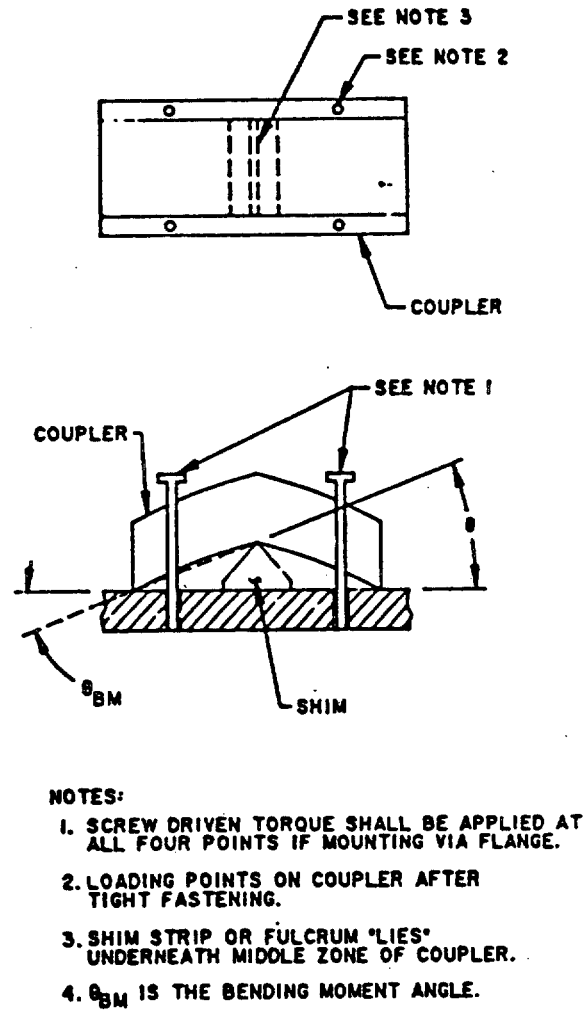
The proper height, Z, of shim required for a fulcrum to induce a 1.15-degree deflecting angle on a coupler of length A is, according to the geometry of figure 3:

$$Z = 1/2\ A \tan(1.15 \text{ degrees})$$
$$Z = 0.01\ A$$



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FIGURE 2. Housing bending moment test, without load.



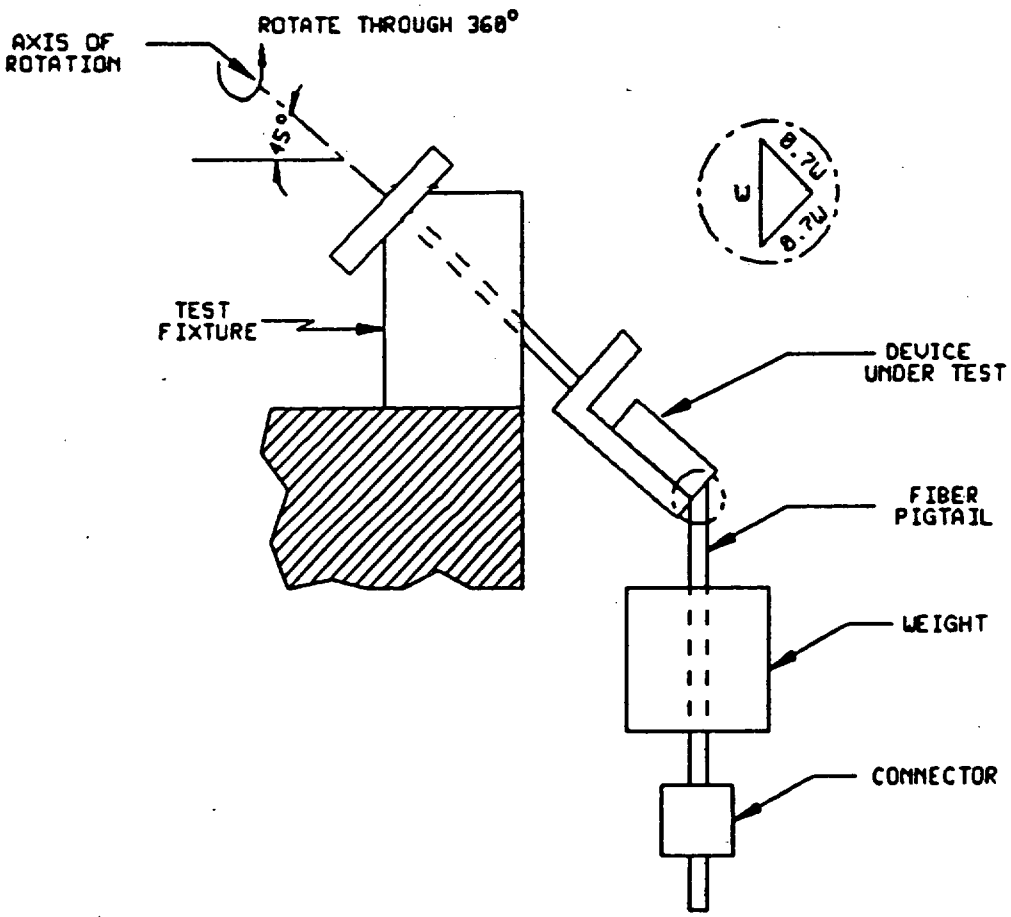
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FIGURE 3. Housing bending moment test, with load.

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4.6.4.6 Twist. Couplers shall be tested in accordance with EIA 455-36 for 50 cycles, with the exception that the test sample shall be a properly pigtailed coupler. The tensile load shall be 5 newtons and the number of loads to be applied shall be one. The change in optical transmittance shall be monitored during and after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.2.6.

4.6.4.7 Fiber dynamic strength. The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied up to the load specified at an angle of 45 degrees to the normal (see figure 4). With the load applied, the pigtail shall be rotated through one rotation (360 degrees). The change in optical transmittance shall be monitored during and after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1. The coupler shall meet the requirements specified in 3.5.2.7.



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FIGURE 4. Fiber dynamic strength test set up.

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4.6.5 Environmental. The following environmental tests shall be performed as specified herein and as specified (see 3.1). The coupler shall include the pigtails and a protective housing.

4.6.5.1 Thermal shock. The coupler shall be tested for 10 cycles in accordance with appendix B, schedule C (see 6.3 and appendix D). The high and low temperatures shall correspond to the specified storage temperatures. The change in optical transmittance shall be monitored after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.1.

4.6.5.2 Fluid immersion. The coupler shall be tested in accordance with EIA 455-12 with the following modifications. Each coupler shall be tested in all of the different immersion fluids. Fluids selected for testing shall be those listed in table VII. At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.2.

TABLE VII. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33 - 37
Turbine fuel (JP-5)	MIL-T-5624	20 - 25
Isopropyl alcohol	TT-I-735	20 - 25
Hydraulic fluids	MIL-H-5606	48 - 50
Lubricating oils	MIL-L-17331 MIL-L-23699	73 - 77
Coolant 1/	-	20 - 25
Sea water	-	20 - 25

1/ Monsanto Coolanol 25, or equal.

4.6.5.3 Temperature life. The coupler shall be tested in accordance with EIA 455-4. The exposure time shall be 96 hours. The exposure temperature shall be 105°C for stand-alone couplers and 125°C for PCB-mounted couplers. The change in optical transmittance shall be monitored after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.3.

4.6.5.4 Temperature-humidity cycling. The coupler shall be tested in accordance with test type 3 of EIA 455-5, except that the sample shall be a coupler, and the temperature extremes shall be the operating extremes as

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specified (see 3.1). The change in optical transmittance shall be monitored during and after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.4.

4.6.5.5 Vibration. The coupler shall be tested in accordance with MIL-STD-1344, method 2005, test conditions II and VI. For test condition VI, test condition C shall apply. Test condition I shall be used during the low frequency portion of test condition II. The change in optical transmittance shall be monitored during and after the test, using equipment with a time resolution of 50 microseconds (μs) (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.5.

4.6.5.6 Shock. Stand-alone couplers shall be tested in accordance with MIL-S-901, grade A, type A, class I. PCB-mounted couplers shall be tested in accordance with MIL-STD-883, method 2002, test condition B. The change in optical transmittance shall be monitored during and after the test using equipment with a time resolution of 50 μs (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.6.

4.6.5.7 Salt spray (corrosion). The coupler shall be tested in accordance with EIA-455-16 for a duration of 96 hours, at a temperature of 35°C. At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.7.

4.6.5.8 Water pressure. The coupler shall be immersed in an aqueous dye penetrant solution at an applied pressure of 0.1 megapascal (equivalent to a depth of 10.4 meters) for 48 hours at a temperature of 10 to 35°C. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the coupler shall be visually examined in accordance with 4.6.1.1 and shall meet the requirements specified in 3.5.3.8.

4.6.5.9 Nuclear radiation resistance. The coupler shall be tested in accordance with appendix C (see 6.3 and appendix D). Unless otherwise specified (see 3.1), tests shall be performed at a wavelength of 1.310 ± 0.020 μm and at the low operating temperature of 20°C, and the high operating temperature. The dose rate shall be 25 rads/sec. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). The coupler shall meet the requirements specified in 3.5.3.9.

4.6.6 Chemical.

4.6.6.1 Fungus resistance. Couplers and materials used in the construction of the couplers that are not identified as fungus-inert shall be tested in accordance with MIL-STD-810, method 508. The couplers shall meet the requirements specified in 3.5.4.1.

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4.6.6.2 Flammability. The coupler shall be tested in accordance with MIL-STD-1344, method 1012, test condition A, except that the test sample shall be a coupler. The change in optical transmittance shall be monitored after the test (see 4.6.3.2). The coupler shall meet the requirements specified in 3.5.4.2.

4.6.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-C-55330.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.10).

5.1 Preservation and packing. The coupler shall be packaged for shipment in accordance with MIL-C-55330. The level of packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The couplers specified herein are intended for coupling and splitting optical signal power. The couplers are suitable for any application within the limitations of the specified performance requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Type, class and style (see 1.2).
(c) Quantity of couplers required.
(d) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) Applicable specification sheet number, title, and date.
(f) When first article is required (see 3.2).
(g) Equivalent test methods, if other than as specified (see 4.6.1.3).
(h) Level of packaging required (see 5.1).
(i) Applicable specification sheet part number (see 6.8).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.6.3.1, 4.6.5.1, 4.6.5.9, and appendix D	DI-MISC-80653	Test reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of eight units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Backscatter. Backscatter is the components of electromagnetic waves that are directed back toward their source when resolved along a line from the source to the point of scatter. In this specification, backscatter is measured as the ratio of the power in dB of the light deflected along a path opposite to the launch path.

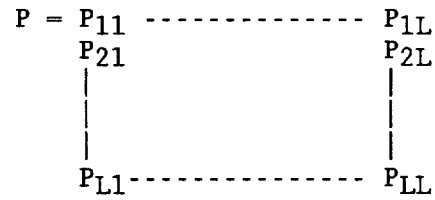
6.6.2 Coupler. A coupler is a passive optical device whose purpose is to distribute or combine optical power among two or more ports.

6.6.3 Coupler transmittance matrix. A coupler transmittance matrix is a matrix of L X L dimensions, where L is the total number of ports (i plus j). Each of the elements of the transmittance matrix is defined by the following equation:

$$P_{ij}\ (dB) = 10 \log_{10} (P_i/P_j)$$

where $P_{ij}(dB)$ is the transmittance in dB from the input port i to the output port j. P_j is the power coming out of port j, and P_i is the power launched into port i. The subscripts of the elements of the transmittance matrix correspond to the port numbers labelled on the coupler. The format of the transmittance matrix is defined as follows:

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6.6.4 Excess loss. Excess loss is fiber optic coupler loss given by the following relation:

$$P_i(\text{excess})\ \text{dB} = \text{SUM}_j\ P_{ij} - \text{SUM}_j\ P_{ij}\ (\text{ideal})$$

where i is the input port, j is the output port, and $P(\text{ideal})_{ij}$ are the transmittance elements which are obtained from an ideal coupler in which all of the input power is coupled to the non-isolated output ports.

6.6.5 Isolation. Isolation is, in a fiber optic coupler, the extent to which optical power from one signal path is prevented from reaching another signal path.

6.6.6 Nonuniformly distributive coupler. A nonuniformly distributive coupler is a fiber optic coupler in which an optical input to each input port of the coupler is not equally distributed among the output ports, and the optical input to each output port of the coupler is not equally distributed among the input ports.

6.6.7 Optical cable pigtail. An optical cable pigtail is a short length of fiber optic cable permanently fixed to a component and is used to couple optical power between it and another optical cable pigtail or fiber optic cable, such as a transmission cable.

6.6.8 Optical fiber pigtail. An optical fiber pigtail is a short length of optical fiber permanently fixed to a component and is used to couple optical power between it and another optical fiber pigtail or a transmission fiber.

6.6.9 Polarization sensitivity. Polarization sensitivity is the change in a performance parameter, such as the power splitting ratio of a coupler, per unit change in the angle of polarization.

6.6.10 70/70 restricted launch. A 70/70 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to 70 percent of the fiber numerical aperture.

6.6.11 70/100 restricted launch. A 70/100 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to or greater than the fiber numerical aperture.

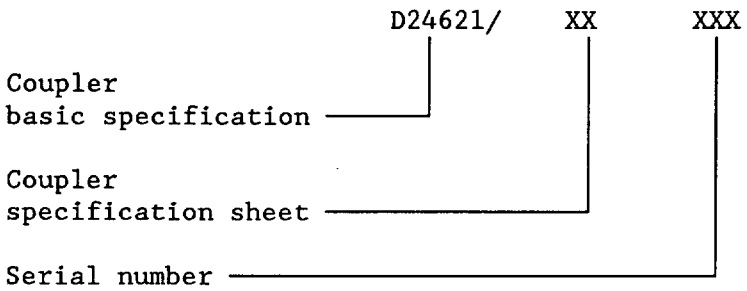
6.6.12 Uniformly distributive coupler. A uniformly distributive coupler is a coupler in which an optical input to each input port of the coupler is equally distributed among the output ports, and an optical input to each output port of the coupler is equally distributed among the input ports.

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6.6.13 Wavelength-division multiplexing (WDM). WDM is transmission in which a wavelength range is divided into narrow bands and each is separately modulated and used as a separate transmission channel.

6.7 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.8 Part or Identifying Number (PIN). Couplers should contain only the following:



Examples:

D24621/01-001
D24621/02-002

6.9 Part designator. A part designator if specified (see 3.1) should include classifications (see 1.2) as follows:

Type (see 1.2.1)
Class (see 1.2.2)
Style (see 1.2.3)

Example: 1A2

6.10 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.11 Subject term (key word) listing.

Ambient light susceptibility
Backscatter
Excess loss
Fiber optic communications
Fiber optic component
Isolation
Multimode
Nonuniformly distributive
Optical combiner
Optical connectorized component
Optical pigtailed device
Optical power combiner
Optical power splitter
Optical splitter
Optical transmittance
Passive coupler
Polarization sensitive
Single mode
Uniformly distributive
Wavelength dependent device
Wavelength division multiplexing
Wavelength independent device
Wavelength sensitive

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6005-N001)

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APPENDIX A

TEST PROCEDURE TO MEASURE RETURN LOSS

10. SCOPE

10.1 Scope. The intent of this procedure is to determine the ratio of optical power reflected by a component or an assembly to the optical power incident on a component port when that component assembly is introduced into a link or system. This ratio is termed return loss. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2 Test. The test for return loss is based on two setups. The first test setup shown on figure 5 is designed to measure the insertion loss of a coupler along a particular through a path of that device. The coupler is an integral part of measuring the insertion loss of the test setup shown on figure 6.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TEST EQUIPMENT

30.1 Apparatus. The apparatus shall consist of the following primary items and those additional items identified on figures 5 and 6 and the specification sheet (see 3.1). The apparatus configuration shall be as shown on figures 5 and 6.

30.1.1 Optical power source. The optical power source shall consist of a light source emitting wavelengths suitable for the devices tested and can either be modulated or unmodulated. A light-emitting diode (LED) or injection laser with suitable radiation band is a suitable source and may be specified. The source shall be stable to plus or minus 1 percent (0.05 dB) over the measurement period. Wavelength selective filters may be used if necessary to limit optical spectral width.

30.1.2 Optical directional coupler. The directional coupler (or splitter) may be of any convenient coupling ratio. Its return path to forward path power ratio (directivity), when terminated in a nonreflecting termination, shall be 10 dB greater than the maximum return loss ratio desired to be measured. The coupler shall be mode insensitive.

30.1.3 Test jumper assemblies. One pair of pigtails are required per device under test (DUT) measurement.

30.1.4 Termination. An optical load which is fully absorbent and nonreflective (at least 10 dB below the desired level of test sensitivity) when coupled to the connecting cable of the test system. Index matching gel or fluid is a suitable termination.

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30.1.5 Optical power meter. A device that measures incident optical power which is linear to within 5 percent over the power levels to be measured and stable to plus or minus 1 percent (0.05 dB) over the measurement period. Relative power levels only are required so no absolute accuracy is necessary.

30.1.6 Interconnects/splices. Since all connectors used in the test will influence the test results, extremely low loss and repeatable units shall be used. Fusion splices are recommended, if practical, and are noted as such on the figures. Index matching materials at all interfaces not under test are highly recommended.

30.1.7 Mode filter. A 50-mm diameter loop of fiber/cable of the same type used in the coupler.

40. TEST SAMPLE

40.1 Test sample. The test sample or DUT shall be as specified (see 3.1).

50. TEST PROCEDURE

50.1 Determination of insertion loss of coupler. The test procedure shall be as follows:

(a) Referring to the test configuration shown in step A of figure 5, a meter of single-mode fiber is fusion-welded to the lead length of the fiber from port 3 of the coupler as a temporary joint. This fiber lead from port 2 is connected to a detector/power meter. Index matching material is applied to fiber leads from ports 1 and 4 to provide termination points. A power measurement is taken at the specified wavelength and a measurement P_2 is recorded.

(b) In step B, cut the fiber between the coupler and the fusion joint, just beyond the joint, allowing material for polishing of the cut end. The free end of the fiber shall be properly prepared so that it is smooth and perpendicular to the axis. Connect the prepared end of the fiber to the detector ensuring that the power level is repeatable. Measure and record this power measurement as P_3. The insertion loss of this particular path between port 2 and port 3 of the coupler is given by the formula:

$$L_c = -10\log_{10} (P_2/P_3)$$

50.2 Determination of DUT return loss. The test procedure shall be as follows:

(a) Using the setup shown on figure 5, connect the fiber lead of port 1 of the coupler to the light source. Connect the detector/power meter to the fiber lead from port 3. Terminate the fiber leads from port 2 and 4 with index matching gel. Measure and record the insertion loss between ports 1 and 3 as P_o.

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(b) Switch the detector/power meter to the fiber lead from port 2 of the coupler and terminate the fiber from port 3 with index gel. Measure and record the insertion loss between ports 1 and 2 as P_i.

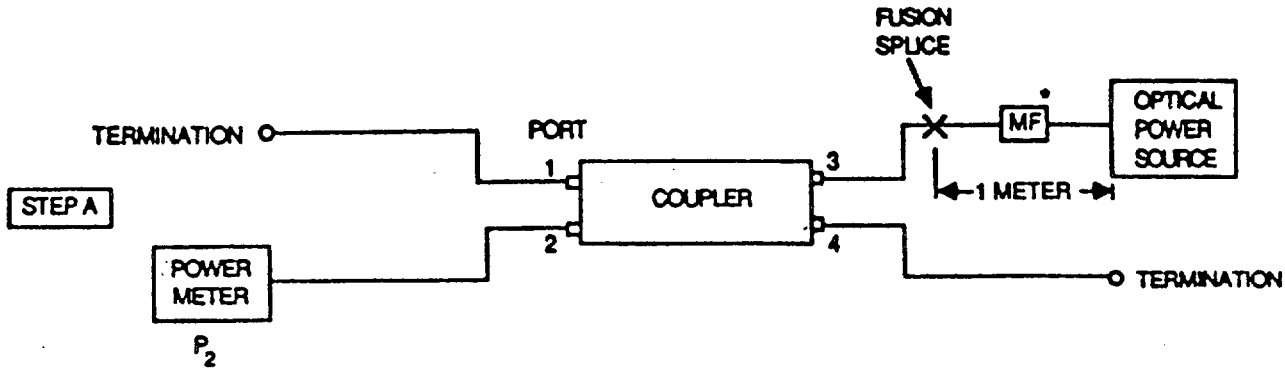
(c) The test jumper assemblies are cut in half to provide the necessary DUT pigtails required for the return loss measurements. Joint the DUT pigtail to the fiber from port 3 with a temporary joint (fusion). The DUT is then coupled to a second pigtail using an adapter. The end of the second pigtail is terminated with index gel. With the light source at the specified wavelength, measure and record the power reflection of the DUT as P'_i. Repeat steps A through C for the remaining DUT's to be measured.

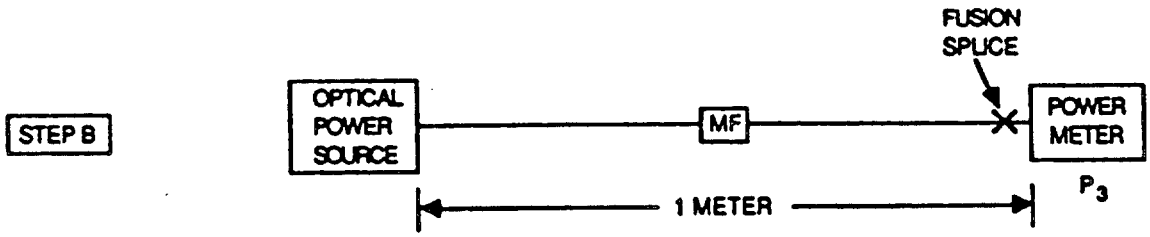
(d) The return loss of each DUT is then calculated by the formula below:

$$R_L = -10\log_{10} \ (P'_i - P_i)/P_o + 10\log_{10} (P_2/P_3)$$

The DUT should have (as an objective) a return loss greater than 30 dB.

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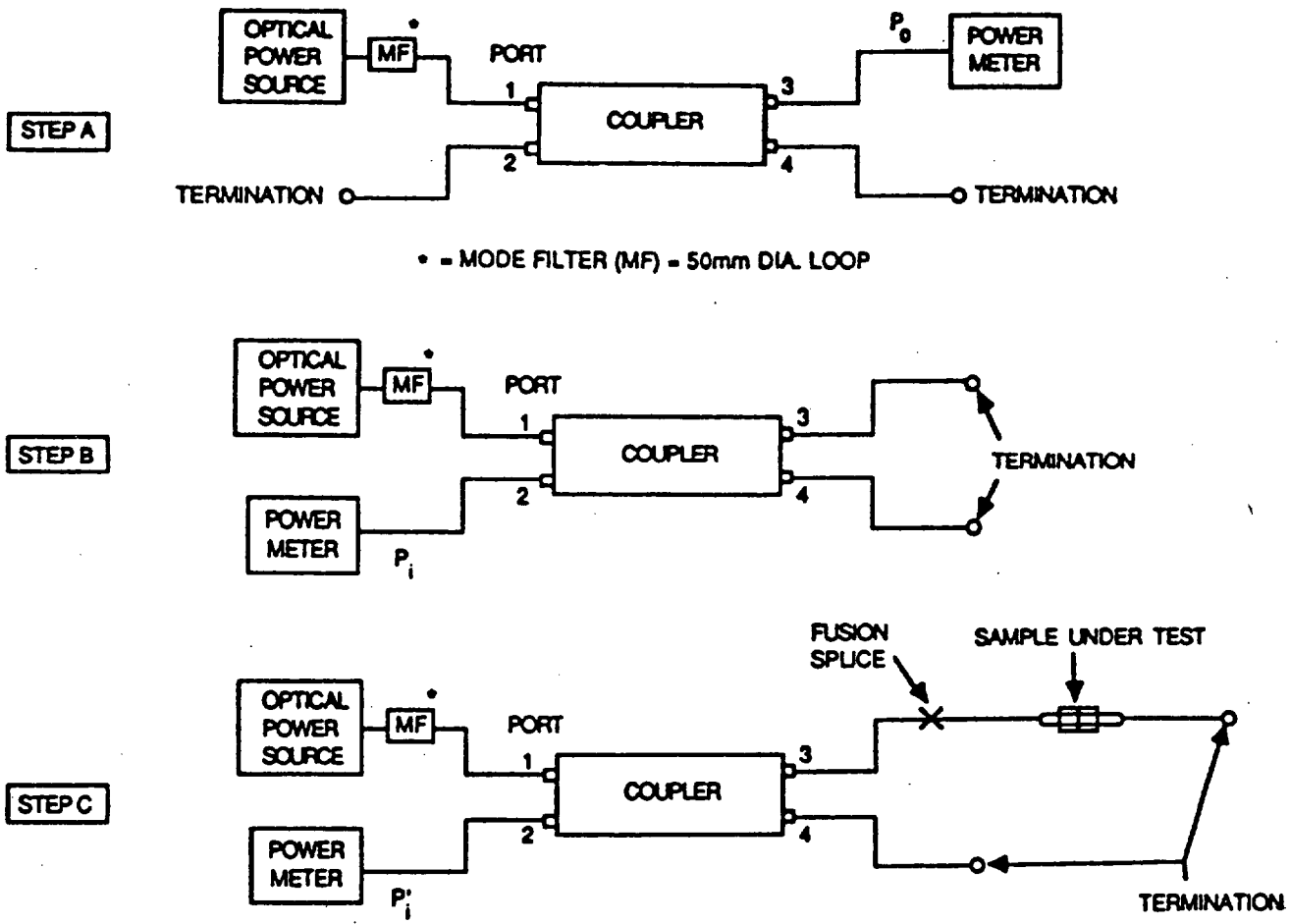




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FIGURE 5. Coupler insertion loss test setup.

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FIGURE 6. DUT return loss test setup.

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APPENDIX B

PROCEDURE TO MEASURE TEMPERATURE SHOCK EFFECTS ON FIBER OPTIC COMPONENTS

10. SCOPE

10.1 Scope. This procedure describes a method for conducting temperature shock testing on optical fiber, cables, cable assemblies, or fiber optic devices. The ability of a fiber optic component to withstand sudden changes in ambient temperature is tested by simulating thermal shock conditions which could arise during shipment, storage, or use. The test for gradual temperature cycling is defined in EIA 455-3. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2 Common damage. Common types of damage or impairment of product function resulting from this test are:

(a) Excessive temporary or permanent changes in transmittance.
(b) Blisters, cracks in materials or accessories, or broken parts.
(c) Inability to mate or unmate, if the sample is a connector.
(d) Degradation of seals or O-rings, if the sample is a connector or device.

20. APPLICABLE DOCUMENTS

20.1 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)
455-57 - Optical Fiber End Preparation and Examination.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

30. TEST EQUIPMENT

30.1 Test equipment. The following test equipment is required for this procedure:

30.1.1 Environmental chambers. Two environmental chambers shall be provided, one for low temperature conditions and the other for high temperature conditions. Alternatively, a single chamber may be used if it has two separate compartments, with a means of transferring from one compartment to the other.

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Each chamber or compartment shall be of sufficient size that the presence of test samples shall not interfere with maintaining a selected temperature within plus or minus 2°C throughout the test. The temperature inside each chamber shall be maintained by air circulation. The chamber layout shall be such that, whenever test samples are in place, air circulation over and around the samples is not obstructed.

30.1.2 Temperature recording devices. Temperature recording devices shall be used to provide a record of each chamber temperature during a test. It may be desirable to have a multichannel recorder connected to thermocouples placed around each test sample. The recorder shall have an accuracy of 1°C across the temperature range of the test.

30.1.3 Transmittance equipment. Equipment shall be provided for measuring the change in transmittance (throughput) in dB of the test sample in accordance with EIA 455-20 or equivalent.

40. TEST SAMPLE

40.1 Test sample. The test sample shall be a representative specimen of the components to be tested.

40.1.1 Sample of optical fiber, cable, or cable assembly. Unless otherwise specified (see 3.1), a sample of optical fiber, cable, or cable assembly shall be of sufficient length to allow at least 500 meters inside the chamber. Fiber test samples shall be deployed in the chamber either as loose coils, or on reels (spools) that provide for minimum tension. Loose coils shall be supported on a horizontal pallet. Fiber samples on a reel (spool) shall be at minimum tension so as to eliminate any possible effects of the differences in thermal expansion and contraction between the sample and the reel. Cable and cable assembly samples shall be loosely wound (minimum tension) on a reel (spool) so as to eliminate any possible effects of the differences in thermal expansion and contraction between the sample and the reel.

40.1.2 Sample of optical connector or device. A sample lot shall be defined as the test sample. The connector or device shall be prepared for testing by installing it on an optical fiber or cable. Installation shall be in accordance with the manufacturer's standard procedures with sealing plugs, and so forth. Connectors or devices not normally equipped with an integral coupling device shall be maintained in the simulated mated condition by means of a suitable fixture. The fixture shall be made as lightweight as possible in order to minimize any "heat sink" effects that could reduce the severity of thermal shock. The length of the support fiber or cable shall be kept to a minimum. Connectors or devices that are part of the test sample within the chamber shall be specified.

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50. TEST PROCEDURE

50.1 Test procedure. The test procedure shall be as follows:

(a) For a fiber or cable sample, the test sample shall be wound on a reel (spool) at a minimum tension, taking care to avoid kinks or loose convolutions. The sample shall be inspected visually to ensure that coils on the reel are uniformly distributed. Alternatively, loose-coiled fiber samples shall be loosely tied with twine at four equally spaced positions, and placed on a horizontal pallet.

(b) The mass of the total test sample, including all parts and test fixtures such as contacts, sealing rings, reels, pallets, connecting cables and mating fixtures, shall be determined before the test begins.

(c) Prior to placing the test sample in the first chamber, the test sample shall be preconditioned at 25 ± 2°C with a relative humidity not greater than 50 percent, for an amount of time not less than that specified in table VIII.

(d) The test chambers shall be stabilized at the chock temperature for a minimum of 15 minutes prior to receiving test samples.

(e) All fiber ends shall be prepared in accordance with the guidelines in EIA 455-57.

(f) The transmittance of the test sample shall be measured at the specified wavelengths. If the sample is a fiber optic cable, optical transmittance of all fiber shall be determined. A transmitting fiber is defined as one that meets or surpasses the optical physical characteristics specified in the specification sheet (see 3.1).

(g) Place the test sample in the first shock chamber with care, avoiding sharp twists or bends in fiber or cable that could give rise to microbending. All samples shall be positioned in the chamber so that sufficient free circulation of air around the sample is provided.

(h) The sample shall be subjected to the temperature conditions of step 1, as specified in the applicable schedule of table IX. The sample shall remain at this temperature for the soak time, corresponding to its mass, as specified in table VIII. (Thermal equilibrium is thereby achieved, while testing time is kept to a minimum.)

(i) Step 2 (see table IX), transfer of the sample to the next chamber, shall be completed within a minimum of 5 minutes. The placement conditions and precautions of 50.1(g) shall be observed.

(j) After transfer, the sample shall be subjected to the temperature conditions of step 3, as specified in the applicable schedule of table IX, for the soak time specified in table VIII.

(k) Step 4 (see table IX) involves the removal of the test sample from the chamber and completes one test cycle.

(l) Steps 1 through 4 (see table IX) shall be repeated for the total number of test cycles as specified (see 3.1). Designations for four test-cycle conditions are given in table X.

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(m) After the specified number of test cycles has been run, the sample shall be allowed to return to room temperature (25 ± 2°C) and remain at room temperature for a minimum of 24 hours.

(n) After completion of the specified number of temperature shock cycles, the transmittance of the test sample shall be measured at the same wavelengths, and by the same method used in 50.1(f) and recorded. If the sample is a multifiber optical cable, the optical transmittance of all fiber shall be determined.

(o) After all optical performance measurements have been made, the sample shall be visually examined for permanent set, cracking, crazing, stickiness, delamination, deterioration of markings, leaking of filling or potting compounds, or other damage.

(p) If the sample has rematable connectors as an integral part, the connectors shall be unmated and remated, and their optical physical performance rechecked and recorded.

TABLE VIII. Mass of test sample verses minimum soak time at temperature.

Test sample mass (kg)	Minimum time at temperature (hours)	
	Preconditioning	Soaking
under 0.35	1	0.5
0.35 to 0.7	2	1
0.70 to 1.50	4	2
1.50 to 15.0	8	4
15.0 to 150.0	16	8
over 150.0	24	12

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TABLE IX. Schedules for temperature shock for one complete cycle.

Schedule	Step	Temperature (°C)		Minimum soak time (minutes)
		Set	Tolerance	
A	1	-55	+0, -3	See table VIII
	2	25	±2	≤ 5
	3	85	+3, -0	See table VIII
	4	25	±2	≤ 5
B	1	Low temperature in accordance with specification sheet		See table VIII
	2	25	±2	≤ 5
	3	High temperature in accordance with specification sheet		See table VIII
	4	25	±2	≤ 5
C	1	High temperature in accordance with specification sheet		See table VIII
	2	25	±2	≤ 5
	3	Low temperature in accordance with specification sheet		See table VIII
	4	25	±2	≤ 5

TABLE X. Designations for number of test cycles.

Designation			Number of complete cycles
Schedule A	Schedule B	Schedule C	
A-0	B-0	C-0	5
A-1	B-1	C-1	25
A-2	B-2	C-2	50
A-3	B-3	C-3	100

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60. CALCULATIONS

60.1 Calculations. Calculations of increase or decrease of transmittance during the entire temperature shock period shall be in accordance with EIA 455-20, or equal.

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APPENDIX C

PROCEDURE FOR MEASURING GAMMA IRRADIATION EFFECTS IN FIBER OPTIC COMPONENTS

10. SCOPE

10.1 Scope. This appendix outlines a method for measuring the steady state response of optical fibers, optical cables, and fiber optic components exposed to gamma radiation. It may be employed to determine the level of radiation-induced attenuation-produced single-mode multimode optical fibers, in either cabled or uncabled form, due to exposure to gamma radiation. This test is not a materials test for the non-optical material components of a fiber optic cable. If degradation of cable materials exposed to irradiation is to be studied, other test methods will be required. This appendix is a mandatory part of this appendix. The information contained herein is intended for compliance.

10.2 Background. The attenuation of cabled and uncabled optical fibers generally increases when exposed to gamma radiation. This is primarily due to the trapping of radiolytic electrons and holes at defect sites in the glass (that is, the formation of color centers). This test procedure focuses on two regimes of commercial interest: the low dose rate regime suitable for estimating the effect of environmental background radiation, and the high dose rate regime suitable for estimating the effect of adverse nuclear environments. The testing of the effects of environmental background radiation is achieved with a two-point attenuation measurement approach similar to EIA 455-46 or EIA 455-78. The effects of adverse nuclear environments are tested by monitoring the power before, during, and after exposure of the test sample to gamma radiation. The depopulation of color centers by light (photobleaching) or by heat causes recovery (lessening of radiation induced attenuation). Recovery may occur over a wide range of time scales ranging from 10^{-2} second to 10^{4} seconds. This complicates the characterization of radiation-induced attenuation since the attenuation depends on the temperature of the test environment, on the total dose, and the dose rate applied to the sample and on the light level used to measure it.

20. APPLICABLE DOCUMENTS

20.1 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-46 - Spectral Attenuation Measurement for Long Length, Graded-Index Optical Fibers. (DoD adopted)

455-78 - Spectral Attenuation Cutback Measurement for Single-Mode Optical Fibers.

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(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

30. TEST EQUIPMENT

30.1 Radiation source.

30.1.1 Testing of environmental background radiation. A cobalt-60, or equal, ionizing source (for example, Cesium 137) shall be used to deliver gamma radiation at a low dose rate of not greater than 20 rads/hour (see figure 7).

30.1.2 Testing of adverse nuclear environments. A cobalt-60, or equal, ionizing source (for example, Cesium 137) shall be used to deliver gamma radiation at a dose rate ranging from 5 rads/second to 250 rads/second (see figure 8).

30.2 Light source. A light source such as a tungsten-halogen lamp or set of lasers or LED's shall be used to produce radiant energy at wavelengths of 1.31 μm. The light source shall be stable in intensity over a time period sufficient to perform the measurement. Unless otherwise specified (see 3.1), the power coupled from the source into the test sample shall be not greater than 30 dBm (1.0 μwatts). The light source shall be modulated with a pulse signal at a 50-percent duty cycle. Note: If a source that couples more than 1.0 μwatt is used, photobleaching may occur.

30.3 Optical filters or monochromators. Unless otherwise specified (see 3.1), wavelengths of 1.300 ± 0.020 μm shall be obtained by filtering the light source with a set of optical filters or a monochromator. The 3-dB optical bandwidth of the filters shall be not greater than 0.025 μm.

30.4 Cladding mode stripper. When applicable, a device that extracts cladding modes shall be employed at the input end of the test sample. If the fiber coating materials are designed to strip cladding modes, a cladding mode stripper is not required.

30.5 Fiber support and positioning apparatus. A means of stably supporting the input end of the test sample, such as a vacuum chuck, shall be arranged. This support shall be mounted on a positioning device so that the end of the test sample can be repeatedly positioned in the input beam.

30.6 Optical splitter. An optical splitter shall divert a small portion of the input light to a reference detector. The reference path shall be used to monitor system fluctuations for the duration of the test.

30.7 Input launch simulator.

30.7.1 Graded-index multimode fiber. An equilibrium mode simulator shall be used to attenuate higher order propagating modes and to establish a steady-state mode condition near the input end of the fiber.

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30.7.2 Single-mode fiber. An optical lens system or fiber pigtail may be employed to excite the test fiber. The power coupled into the test sample shall be stable for the duration of the test. If an optical lens system is used, a method of making the positioning of the fiber less sensitive is to overfill the fiber end spatially and angularly. If a pigtail is used, it may be necessary to use index matching material to eliminate interference effects. A high order mode filter shall be employed to remove high order propagating modes in the wavelength range not less than the cutoff wavelength of the test fiber.

30.8 Detector - signal detection electronics. An optical detector which is linear and stable over the range of intensities that are encountered shall be used. A typical system might include a photovoltaic mode photodiode amplified by a current input preamplifier, with synchronous detection by a lock-in amplifier.

30.9 Optical power meter. An optical power meter shall be used to determine that the power coupled into the optical source into the test sample is not greater than 1.0 μwatt or as specified (see 3.1).

30.10 Radiation dosimeter. Thermoluminescent LiF or CaF Crystal Detectors (TLD's) shall be used to measure the total radiation dose received by the specimen fiber.

30.11 Temperature-controlled container. Unless otherwise specified (see 3.1), the temperature-controlled container shall have the capability of maintaining the specified temperatures to within plus or minus 2°C.

30.12 Test reel. The test reel shall not act as a shield or sink for the radiation used in this test.

40. TEST SAMPLE

40.1 Specimens.

40.1.1 Fiber specimen. The test specimen shall be a representative sample of the fiber as specified (see 3.1).

40.1.2 Cable specimen. The test specimen shall be a representative sample of the cable as specified (see 3.1) and shall contain at least one of the specified fibers.

40.1.3 Other specimens. The test specimen shall be a fiber optic component with a fiber or cable pigtail as specified (see 3.1).

40.2 Specimen for environmental background radiation test. Unless otherwise specified (see 3.1), the length of the test sample shall be 3 kilometers (km) plus or minus 20 meters. A minimum length at the ends of the test sample (typically not greater than 5 meters) shall reside outside of the test chamber and shall be used to connect the optical source to the detector. The irradiated length of the test sample shall be reported.

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40.3 Specimen for testing adverse nuclear environments. Unless otherwise specified (see 3.1), the length of the test sample shall be 50 ± 1 meters. A minimum length at the ends of the test sample (typically not greater than 5 meters) shall reside outside the test chamber and be used to connect the optical source to the detector.

40.4 Test reel. The test sample shall be spooled onto a reel with a drum diameter as specified (see 3.1). Allowance shall be made for the unspooling of a measured length of the test sample from each end of the reel to allow for attachment to the optical measurement equipment.

40.5 Ambient light shielding. The test sample shall be shielded from ambient light to prevent external photobleaching.

50. TEST PROCEDURE

50.1 Calibration of radiation source. Calibration of the radiation source for dose uniformity and level shall be made prior to the test sample being set up in the chamber. Four TLD's shall be placed where the axis of the test reel will be placed. (Four TLD's are used to get a representative average value.) A dose not less than the actual test dose shall be used to calibrate the system. To maintain the highest possible accuracy in measuring the test dose, the TLD's shall not be used more than once.

50.2 Fiber end preparation. The test sample shall be prepared such that its endfaces are smooth and perpendicular to the fiber axis as specified in EIA 455-57.

50.3 Environmental background radiation test. The procedures for measuring the attenuation (two point loss measurement) of the test sample before and after exposure to the gamma radiation source are described below.

(a) The reel of fiber or cable to be tested shall be placed in the test set-up as shown on figure 7.
(b) The input end of the fiber shall be placed in a positioning device and aligned. The output end shall be positioned so that all light exiting the fiber impinges on the active surface of the detector.
(c) The test sample shall be preconditioned in the temperature chamber at 25 ± 5°C for 1 hour prior to testing, or at the test temperature for a preconditioned time as specified (see 3.1).
(d) A two-point attenuation measurement of the test sample shall be performed at the specified test wavelengths in accordance with EIA 455-46 or EIA 455-78. The attenuation, A, of the fiber prior to exposure to the gamma radiation source shall be measured.
(e) The power at the input end of the test sample (point A on figure 7) shall be measured with a calibrated power meter. If necessary, the source level shall be adjusted so that the power at point A is less than 1.0 μwatt or as specified (see 3.1).
(f) The test sample ends shall be prepared in accordance with 50.2 and aligned in the test set as specified in 50.3(b).

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(g) With the radiation source off, the input end of the test sample shall be positioned to obtain maximum optical power at the detector. Once set, the input launch conditions shall not be changed during the gamma irradiation portion of the test.
(h) Prior to irradiation, the output power shall be measured at all test wavelengths at the specified test temperature.
(i) A chart recorder or continuous measurement device shall be connected to the detection system so that a continuous power measurement can be made. The measurement equipment shall be set up such that the detection signal does not exceed the limits of the equipment.
(j) Environmental background radiation effects due to exposure to gamma irradiation shall be determined by subjecting the test sample to dose rates not greater than 20 rads/hour. The test sample shall be exposed to a minimum total dose of 100 rads.
(k) The output power from the test sample shall be measured for the duration of the gamma irradiation cycle.
(l) Upon completion and within 2 hours of the irradiation process, a two-point attenuation measurement of the test sample shall be performed as specified in 50.3(d). The attenuation, A2, of the test sample after exposure to the gamma radiation source shall be measured.

50.4 Adverse nuclear environment test. The procedures for measuring the power propagating in the test sample before, during, and after exposure to the gamma radiation source are described below.

(a) The ends of a short length of test sample (1 to 2 meters) shall be prepared as specified in 50.2.
(b) The input end of the short test length shall be placed in the positioning device and aligned in the test set (see figure 8) to obtain maximum optical power as measured with a calibrated power meter. If necessary, the source level shall be adjusted, using neutral density filters to obtain an optical power level the output of the short length of test sample that is less than 1.0 μwatt or as specified (see 3.1). Note: If a source that couples more than 1.0 μwatt is used, photobleaching may occur.
(c) The test sample reel shall be placed in the test set-up as specified in figure 8.
(d) The input end of the test sample shall be placed in a positioning device and aligned. The output end shall be positioned so that all light exiting the test sample impinges on the active surface of the detector.
(e) The test sample shall be preconditioned in the temperature chamber at 25 ± 5°C for 1 hour prior to testing, or at the test temperature for a preconditioning time as specified (see 3.1).
(f) With the radiation source off, the input end of the test sample shall be positioned to obtain optical power at the detector. Once set, the input launch conditions shall not be changed during the gamma irradiation portion of the test.

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(g) A chart recorder or continuous measurement device shall be connected to the detection system so that a continuous power measurement can be made. The measurement equipment shall be set up so that the detection signal does not exceed the limits of the equipment.

(h) Adverse effects due to exposure to gamma radiation shall be determined by subjecting the test sample to at least one of the dose rates and the total dose level combinations specified in table XI or as specified (see 3.1).

TABLE XI. Total dose and dose rate combinations.

Total dose (rads(Si))	Dose rate (rads/sec)
3,000	5
10,000	50
100,000	200
1,000,000	200

Dose rate levels are only approximate levels since the radiation source characteristics change. A variation in dose as high as plus or minus 50 percent can be expected between sources. The time required to turn the radiation source on or off shall be not greater than 10 percent of the total exposure time.

(i) The output power from the test sample shall be measured for the duration of the irradiation cycle. The power shall also be measured for at least 15 minutes after completion of the irradiation process or as specified (see 3.1). The power level of the reference detector shall also be measured during the recovery time after completion of the irradiation process.

(j) Repeat steps (b) through (i) for the required test temperatures and wavelengths. It will be necessary to use a new non-irradiated specimen for each temperature required.

60. CALCULATIONS

60.1 Calculations. Calculations shall be as follows:

(a) The change in optical attenuation, ΔA (environmental background radiation test), shall be calculated as follows:

$$\Delta A = A2 - A1$$

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(b) The change in optical transmittance, ΔA (testing of adverse nuclear environment), shall be calculated for each wavelength by using the following formula:

$$A_o = -10\log(P_o/P_B) \quad dB$$

$$A_{15} = -10\log_{10}(P_{15}/P_B) \quad dB$$

where:

P is the power output of the test sample immediately after irradiation

P_{15} is the power output of the test sample 15 minutes after irradiation

P_B is the power output of the test sample 15 minutes after irradiation

(c) The results of the reference measurements shall be used to normalize the test results if significant system instability is noted as calculated as in the following formula:

$$A_{REF} = -10\log(P_{E'}/P_{B'}) \quad dB$$

where:

$P_{E'}$ is the power measured by the reference detector at the end of the measurement.

$P_{B'}$ is the power measured by the reference detector at the beginning of the measurement.

(d) Normalized test results that account for system instability are calculated with the following formula:

$$A_{ONOR} = A_o - A_{REF}$$

$$A_{15NOR} = A_{15} - A_{REF}$$

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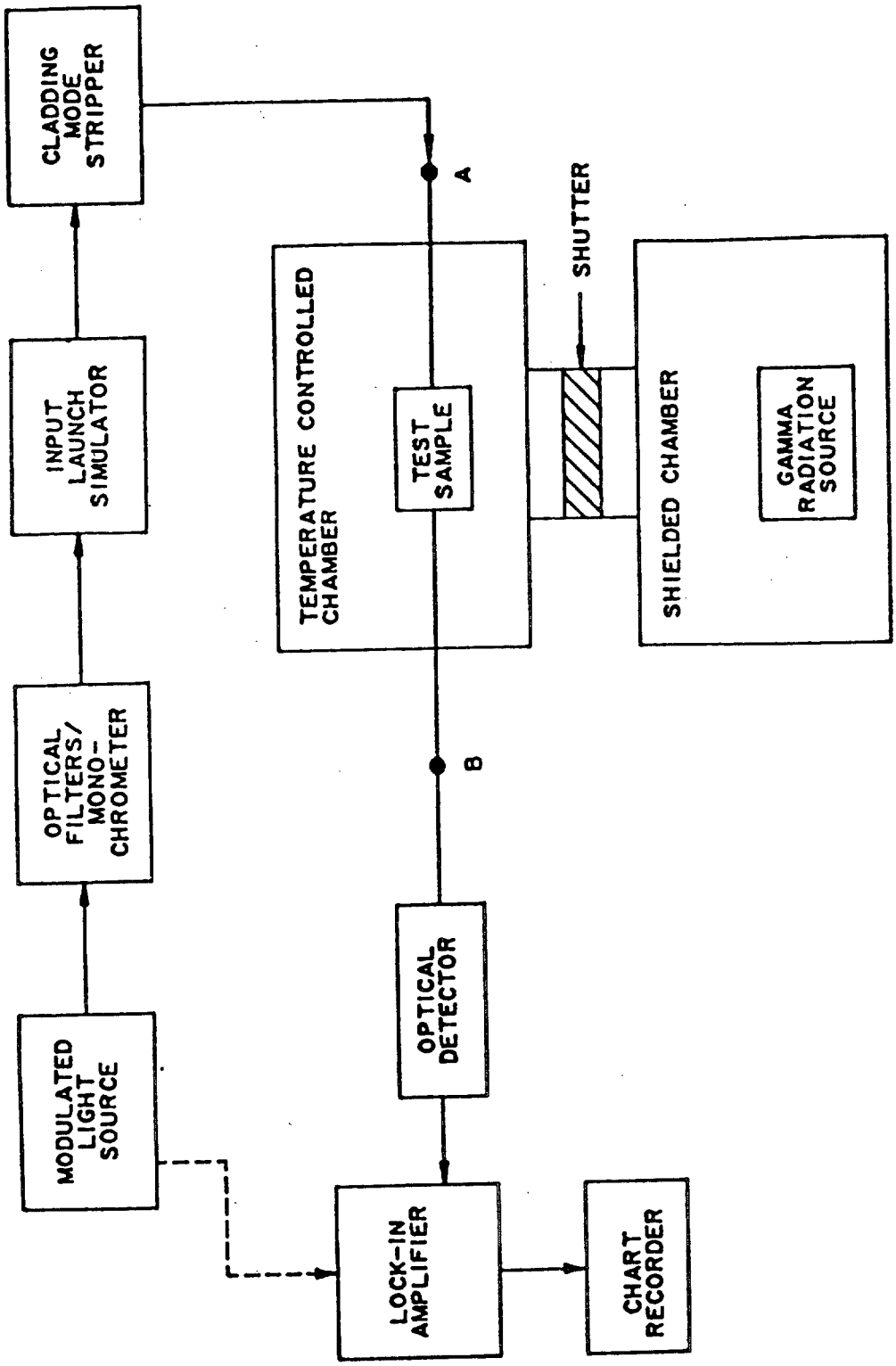


FIGURE 7. Instrumentation for environmental background radiation testing.

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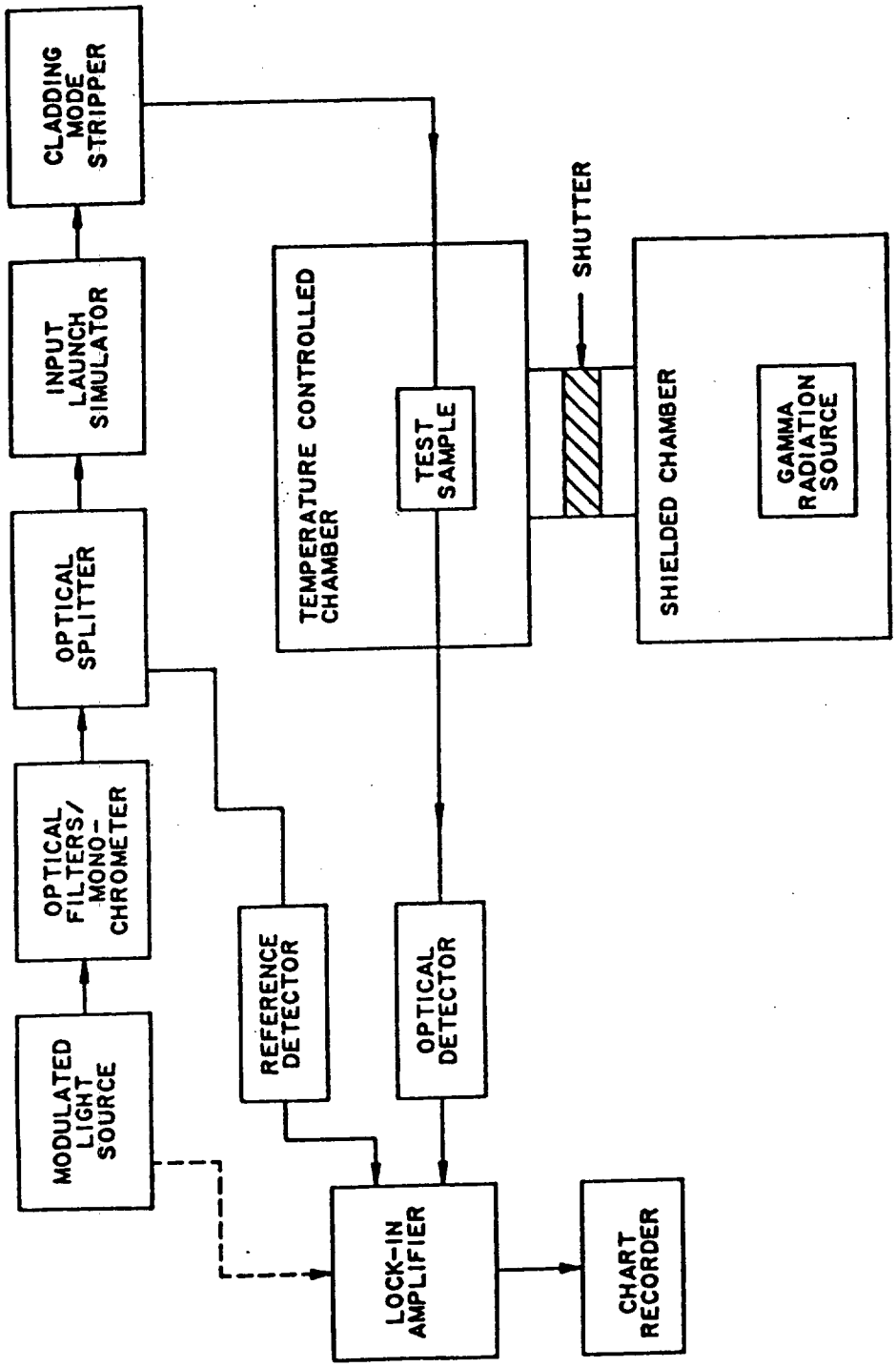


FIGURE 8. Instrumentation for testing the effects of adverse nuclear environments.

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APPENDIX D

TEST REPORTS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the information that shall be included on test reports when required by the contract or order. This appendix is applicable only when data item description DI-MISC-80653 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TEST REPORTS

30.1 Optical transmittance return loss test report. When required by the contract or order, the return loss test report (see 4.6.3.1 and appendix A) shall include the following information:

(a) Title of the test.
(b) Test sample description.
(c) Test equipment used and calibration data.
(d) Launch conditions.
(e) Test values and results.
(f) Applicable specification sheet reference.

30.2 Thermal shock effects test report. When required by the contract or order, the thermal shock effects test report (see 4.6.5.1 and appendix B) shall include the following information:

(a) Date and title of test.
(b) Name of test operator.
(c) Identification of the test samples, including test fixtures, if applicable; the mass of the sample and its length and number of fibers, if applicable; the deployment method for fiber, cable, and cable assemblies.
(d) Identification of the thermal shock schedule used and the number of cycles completed.
(e) Results of visual examination.
(f) Test results, including increase or decrease in transmittance. In the case of a multifiber test cable or assembly, results for each fiber shall be reported.
(g) Equipment documentation, as follows:

(1) Description of the test equipment used and the date of the last calibration. Data shall include wavelengths used, launch conditions, and the uncertainty of error (1 sigma) in the transmittance measurement system.

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(2) If the sample is a fiber, cable, or cable assembly, a description of the coil configuration or the reel (spool).
(3) If the sample is a connector or device and requires a fixture to maintain mateability, a description of the fixtures used in the test.

30.3 Gamma irradiation effects test report. When required by the contract or order, the gamma irradiation test report (see 4.6.5.9 and appendix C) shall include the following information:

(a) Date of test.
(b) Title of test.
(c) Length of test sample exposed to irradiation.
(d) Test wavelengths.
(e) Test temperatures.
(f) Change in attenuation ΔA (environmental background radiation test).
(g) Change in optical transmittance A_o, A_{15} (adverse nuclear environment).
(h) Characteristics of test sample such as fiber type, cable type, dimensions and configuration.
(i) Chart recording of test events.
(j) The following test equipment information:

(1) Description of radiation source.
(2) Description of dosimeters used.
(3) Type of optical source.
(4) Description of optical filters or monochromator.
(5) Description of cladding mode stripper.
(6) Description of input launch simulator and launch conditions used.
(7) Type of optical splitter used.
(8) Description of detection and recording apparatus.
(9) Description of the characteristics of temperature chamber.
(10) Date of latest calibration of test equipment.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-0024621A(NAVY)	COUPLERS, PASSIVE, FIBER OPTIC, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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